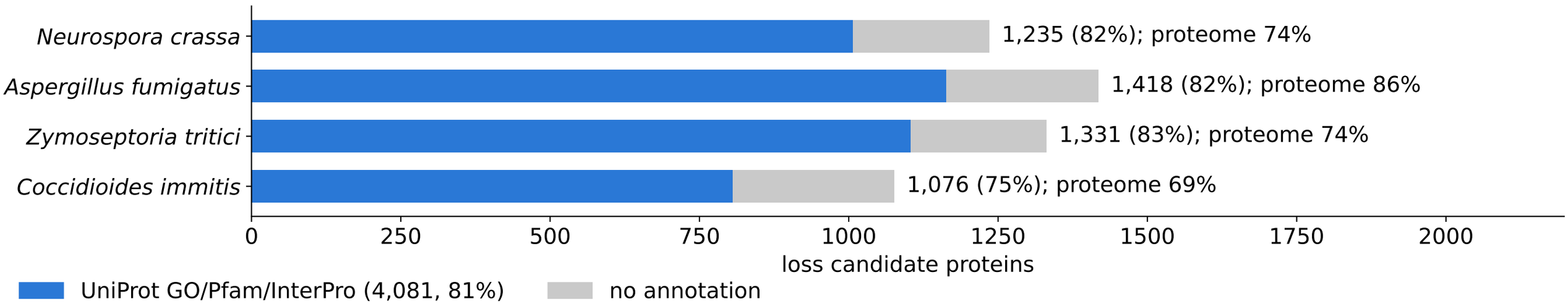
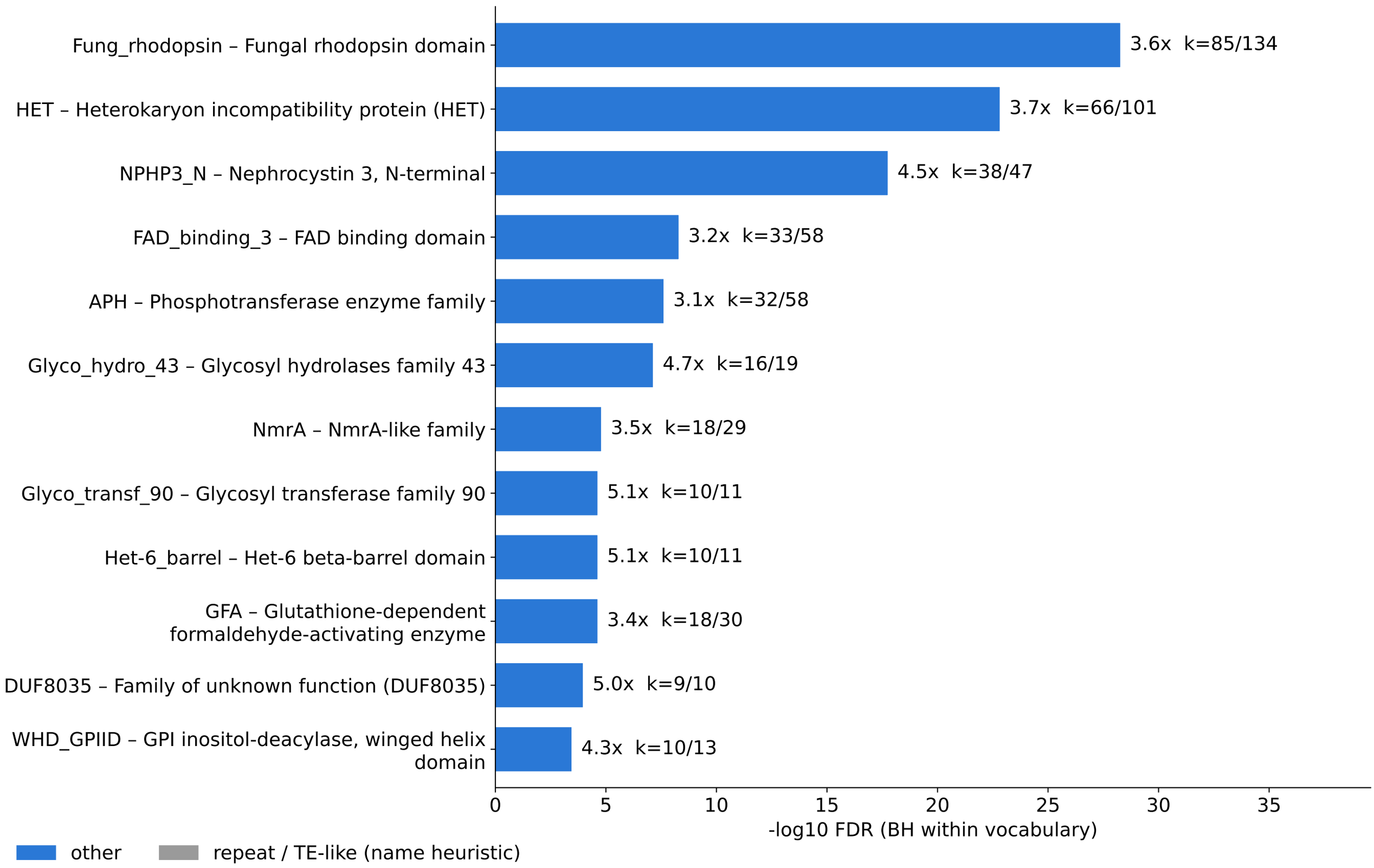
filamentous Ascomycota (4 OUT) vs yeasts (4 IN): enrichment of loss-direction candidates (present in >=0.75 of OUT, absent from all IN; diamond pairwise, nf_NovInvenio main d03f5a2 run 2026-10-13)

A 5,060 candidates in 4 OUT species



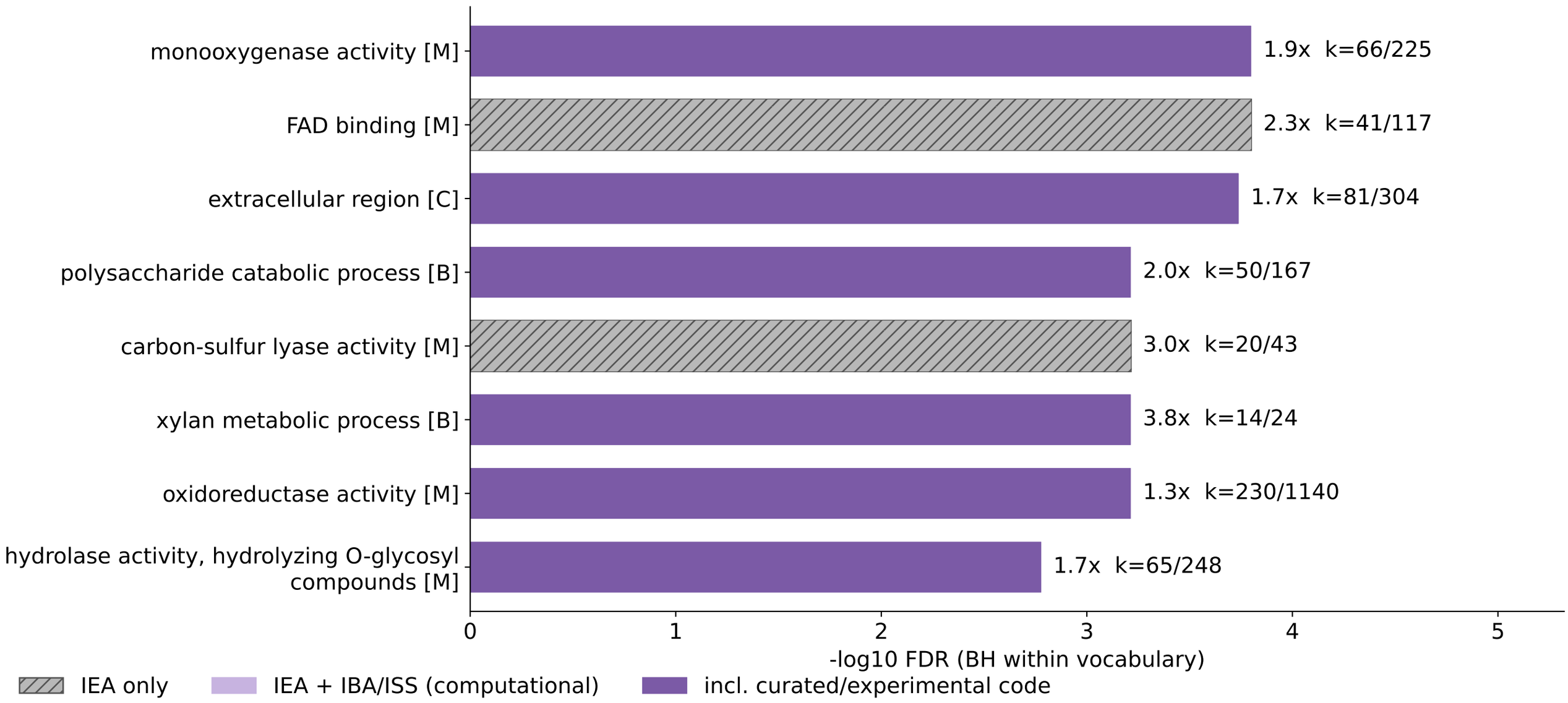
B Pfam families enriched in candidates (primary; family units on both sides)

778 terms tested; 39 at FDR<0.05; annotated units: 2184 cand. / 12252 all



C GO terms (secondary; UniProt computational evidence)

3477 terms tested; 45 at FDR<0.05; annotated units: 1670 cand. / 10932 all



Annotation: UniProt reference proteomes (Pfam, InterPro, GO). Universe: all proteins of the 4 OUT proteomes. Family units: loss-candidate families from the pipeline (mmseqs on loss_candidates.fa); background clustered with mmseqs (0.3 id, 0.8 cov, cov-mode 0). One-sided hypergeometric; BH within each vocabulary. Bars: FDR<0.05 and k>=3 candidate units. Loss candidates: 5060; annotated fraction 0.8065.